

SOFTWARE REQUIREMENT SPECIFICATION

Quiz Application (ANG.M.A001.Opt1)

Group 4

– Hanoi, Oct 2025 –

Table of Contents

I. Record of Changes 1

II. Software Requirement Specification 2

1. Overall Requirements 2

1.1 Context Diagram 2

1.2 Main Business Processes 3

1.3 User Requirements 4

1.4 Database Analysis and Design 7

2. Use Case Specifications 11

2.1 UC01 – View Quiz List 11

2.2 UC02 – View Quiz Details 12

2.3 UC03 – Create Quiz 12

2.4 UC04 – Update Quiz 13

2.5 UC05 – Delete Quiz 14

2.6 UC06 – Manage Questions 14

2.7 UC07 – Manage Answers 15

2.8 UC08 – View Available Quizzes 16

2.9 UC09 – View Quiz Details 17

2.10 UC10 – Take Quiz 18

I. Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Overall Requirements

1.1 Context Diagram

The Quiz Application is an internal training assessment system designed to support knowledge assessment activities within a corporate training environment.

The system enables Trainers to create and manage quizzes, including their questions and answers. Trainees can access available quizzes, answer the questions, and submit their responses for assessment.

The initial release focuses on quiz management and quiz participation. Advanced training management capabilities, such as assigning quizzes to individual Trainees, maintaining assessment history, reporting, and analytics, are outside the scope of this release.

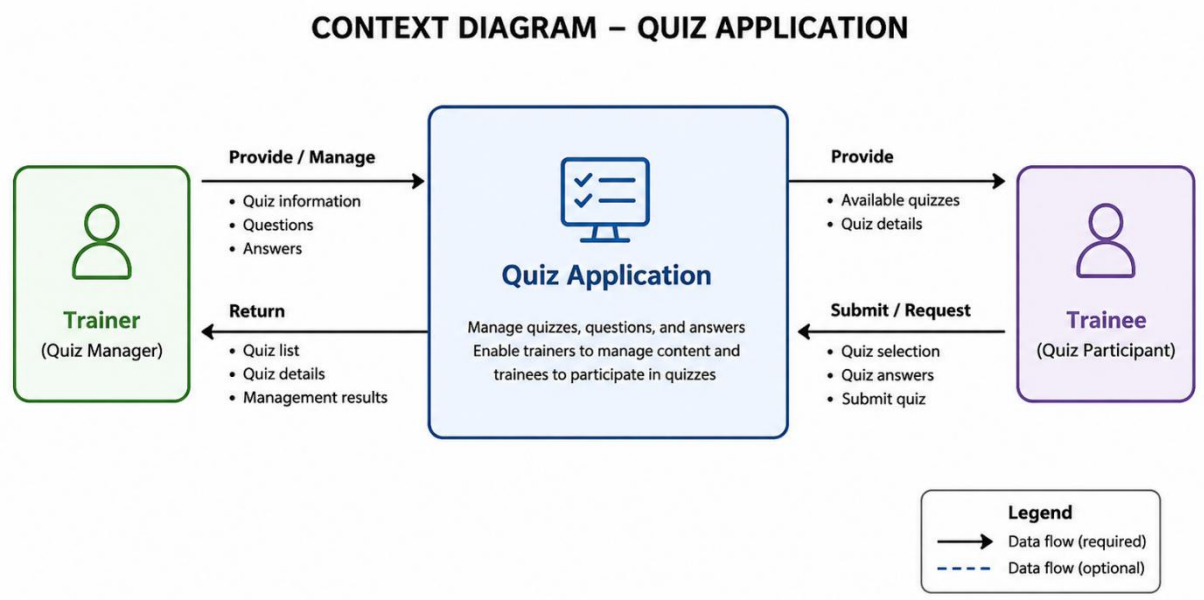


Figure 1. Quiz Application Context Diagram

The context diagram illustrates the system boundary and the interactions between the Quiz Application and its external actors. Trainers provide and maintain quiz-related information, while Trainees access available quizzes and submit their answers through the system.

1.2 Main Business Processes

The Quiz Application supports two main business processes: Quiz Management and Quiz Participation.

The Quiz Management process enables Trainers to maintain the quiz content used for internal training assessments. The Quiz Participation process enables Trainees to access available quizzes, answer questions, and submit their responses.

These processes represent the primary interactions between the system and its users within the scope of the initial release.

1.2.1 Quiz Management Process

The Quiz Management process enables a Trainer to create and maintain quizzes used for training assessments. A quiz consists of general quiz information, questions, and their corresponding answers.

The process begins when the Trainer accesses quiz management and selects an appropriate management action. Depending on the selected action, the Trainer can create a new quiz, modify an existing quiz, or delete an existing quiz.

When creating or updating a quiz, the Trainer provides or modifies the quiz information and manages its associated questions and answers. The system validates the provided data before applying the requested changes. The relationship between a quiz, its questions, and their answers must remain consistent throughout the operation.

If the provided data is valid, the system applies the changes and informs the Trainer that the operation has been completed successfully. If the data is invalid or the requested quiz cannot be found, the system rejects the operation and provides an appropriate error response.

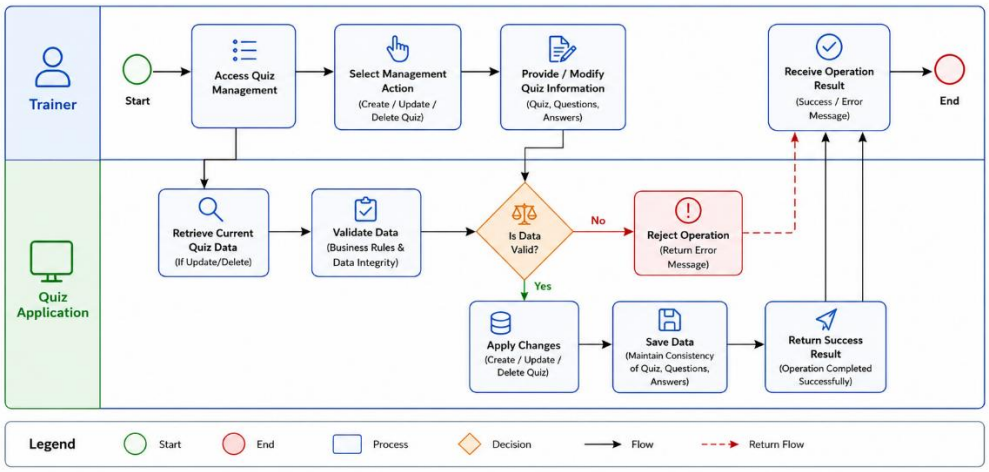


Figure 2. Quiz Management Business Process

1.2.2 Quiz Participation Process

The Quiz Participation process enables a Trainee to participate in an available quiz as part of an internal training assessment.

The process begins when the Trainee accesses the available quizzes and selects a quiz to participate in. The system retrieves and presents the selected quiz together with its questions and the information required for the Trainee to complete the quiz.

The Trainee provides answers to the quiz questions and submits the completed quiz. The system validates the submitted information to ensure that the submission can be processed.

If the submission is valid, the system accepts the Trainee's responses and confirms that the quiz has been submitted successfully. If the submission is invalid, the system rejects the submission and provides an appropriate error response so that the Trainee can correct the relevant information when applicable.

The initial release does not include quiz assignment to individual Trainees, assessment history, reporting, or analytics.

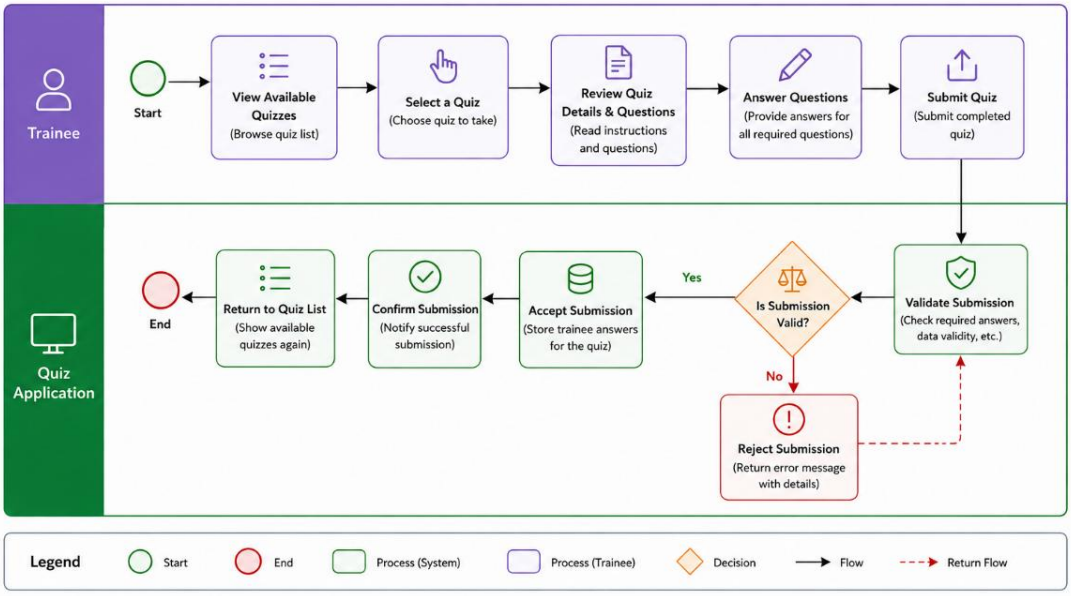


Figure 3. Quiz Participation Business Process

1.3 User Requirements

This section identifies the actors that interact with the Quiz Application and the use cases that represent their primary interactions with the system.

1.3.1 Actors

The Quiz Application has two primary actors: Trainer and Trainee.

No.	Actor	Description
1	**Trainer**	A person responsible for preparing and maintaining quiz content for internal training assessments. The Trainer manages quizzes and their associated questions and answers.
2	**Trainee**	A person participating in the company's training activities. The Trainee accesses available quizzes, answers quiz questions, and submits completed quizzes for assessment.

1.3.2 Use Cases (UC)

The following use cases describe the primary interactions between the actors and the Quiz Application within the scope of the initial release.

ID	Use Case	Feature	Use Case Description
UC01	View Quiz List	Quiz Management	Trainer views the list of existing quizzes available in the system.
UC02	View Quiz Details	Quiz Management	Trainer views the details of a selected quiz, including its questions and answers.
UC03	Create Quiz	Quiz Management	Trainer creates a new quiz with the required quiz information.
UC04	Update Quiz	Quiz Management	Trainer modifies the information of an existing quiz.
UC05	Delete Quiz	Quiz Management	Trainer removes an existing quiz from the system.
UC06	Manage Questions	Question Management	Trainer adds, updates, or removes questions associated with a quiz.
UC07	Manage Answers	Answer Management	Trainer adds, updates, or removes answers associated with a question and identifies the correct answer where applicable.
UC08	View Available Quizzes	Quiz Participation	Trainee views quizzes that are available for participation.
UC09	View Quiz Details	Quiz Participation	Trainee views the information of a selected quiz before starting the quiz.
UC10	Take Quiz	Quiz Participation	Trainee participates in a selected quiz by answering its questions and submitting the completed responses to the system.

1.3.3 Use Case Diagrams

The following use case diagrams illustrate the interactions between the primary actors and the Quiz Application. Separate diagrams are provided for the Trainer and Trainee to clearly represent the responsibilities and system capabilities available to each actor.

1.3.3.1 Use Cases for Trainer

The Trainer is responsible for preparing and maintaining quiz content used in internal training assessments. The Trainer can view existing quizzes, create new quizzes, update or delete existing quizzes, and manage the questions and answers associated with each quiz.

The Trainer interacts with the following use cases:

- UC01 – View Quiz List
- UC02 – View Quiz Details
- UC03 – Create Quiz
- UC04 – Update Quiz

- UC05 – Delete Quiz
- UC06 – Manage Questions
- UC07 – Manage Answers

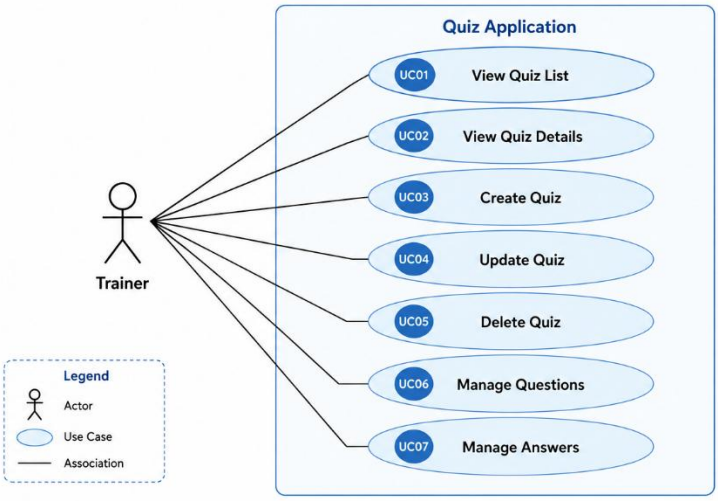


Figure 4. Trainer Use Case Diagram

1.3.3.2 Use Cases for Trainee

The Trainee participates in quizzes provided as part of the company's internal training activities. The Trainee can view available quizzes, review the general information of a selected quiz, and take the quiz by answering its questions and submitting the completed responses.

The Trainee interacts with the following use cases:

- UC08 – View Available Quizzes
- UC09 – View Quiz Details
- UC10 – Take Quiz

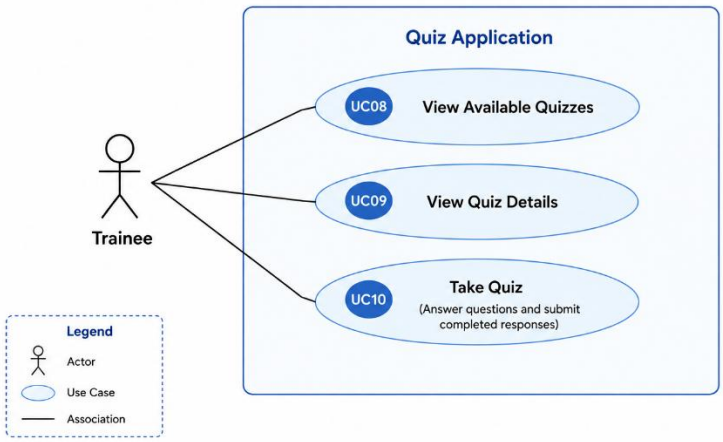


Figure 5. Trainee Use Case Diagram

1.4 Database Analysis and Design

The database of the Quiz Application is designed to support the two main business areas defined within the current system scope: Quiz Management and Quiz Participation.

The database maintains the structure and content of quizzes, including quiz information, questions, and predefined answers. It also provides the basic data structure required to represent quiz submissions and the responses submitted for individual questions during quiz participation.

The database design focuses only on the data required by the current system scope. Advanced training management features such as user authentication, quiz assignment, scoring, assessment history, reporting, and analytics are not included in the current database design.

1.4.1 Database Analysis

Based on the business processes and use cases defined in the previous sections, the data requirements of the Quiz Application are divided into two main areas: Quiz Management Data and Quiz Participation Data.

Quiz Management Data

The Quiz Management process requires the system to maintain the structure and content of each quiz.

A Quiz stores the general information required to describe an assessment. Each Quiz may contain multiple Questions, while each Question belongs to one Quiz.

A Question stores the content of an assessment item and its question type. The question type determines how the question is presented and answered. The supported question types are:

- Multiple Choice
- Single Choice
- True/False
- Fill in the Blanks
- Short Answer
- Long Answer

Depending on its question type, a Question may contain multiple predefined Answers. An Answer stores the answer content and indicates whether the answer is considered correct where applicable.

The relationships between Quiz, Question, and Answer must remain consistent when quiz content is created, updated, or deleted.

** Quiz Participation Data*

The Quiz Participation process requires the system to represent a completed quiz submission and the responses provided for individual questions.

A QuizSubmission represents a submitted attempt for a specific Quiz. Each Quiz may be associated with multiple QuizSubmissions, while each QuizSubmission belongs to one Quiz.

A SubmissionAnswer represents a response provided for an individual Question within a QuizSubmission. Each SubmissionAnswer belongs to one QuizSubmission and refers to one Question.

Depending on the question type, a SubmissionAnswer may reference a predefined Answer or store a textual response. This structure allows the system to represent responses for both predefined-answer questions and text-based questions without introducing scoring or assessment-history functionality.

Based on these data requirements, the following five main entities are identified:

#	Entity	Purpose
1	Quiz	Represents the general information of a quiz used for internal training assessment.
2	Question	Represents a question belonging to a quiz and identifies its question type.
3	Answer	Represents a predefined answer associated with a question.
4	QuizSubmission	Represents a submitted attempt for a quiz during quiz participation.
5	SubmissionAnswer	Represents a response provided for an individual question within a quiz submission.

The main relationships among the identified entities are as follows:

- One Quiz may contain zero or more Questions, while each Question belongs to one Quiz.
- One Question may contain zero or more predefined Answers, while each Answer belongs to one Question.
- One Quiz may be associated with zero or more QuizSubmissions, while each QuizSubmission belongs to one Quiz.
- One QuizSubmission may contain multiple SubmissionAnswers, while each SubmissionAnswer belongs to one QuizSubmission.
- One Question may be referenced by multiple SubmissionAnswers, while each SubmissionAnswer refers to one Question.

1.4.2 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure of the Quiz Application database and the relationships among the entities identified during database analysis.

The Quiz, Question, and Answer entities form the core structure used to manage quiz content. The QuizSubmission and SubmissionAnswer entities extend the data model to support basic quiz participation and submission.

Primary keys are used to uniquely identify individual records, while foreign keys are used to establish relationships between related entities and maintain referential integrity.

The ERD uses Crow's Foot notation to represent the cardinality of the relationships among the entities.

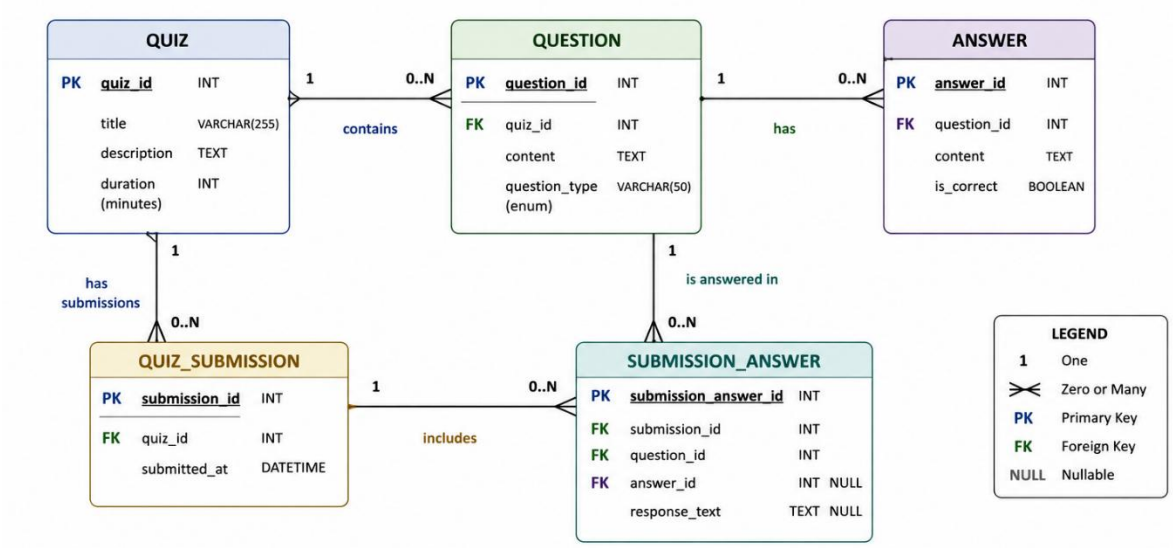


Figure 6. Quiz Application Entity Relationship Diagram

** Entities Description*

#	Entity	Description
1	Quiz	Stores the general information of a quiz used for internal training assessment.
2	Question	Stores the questions associated with a quiz and identifies the type of each question.
3	Answer	Stores the predefined answers associated with a question and identifies the correct answer where applicable.
4	QuizSubmission	Stores information about a submitted attempt for a quiz during quiz participation.
5	SubmissionAnswer	Stores the response provided for an individual question within a quiz submission.

1.4.3 Database Table Design

The following tables define the logical database structure derived from the entities and relationships identified above. Each table specifies the required attributes, data types, constraints, and purpose of the stored data.

** Quiz*

The Quiz table stores the general information of each quiz.

Column	Data Type	Constraints	Description
quiz_id	INT	PK, Auto Increment	Unique identifier of the quiz.
title	VARCHAR(255)	NOT NULL	Title of the quiz.

Column	Data Type	Constraints	Description
description	TEXT	NULL	Description of the quiz.
duration	INT	NOT NULL	Duration of the quiz in minutes.

** Question*

The Question table stores the questions associated with quizzes.

Column	Data Type	Constraints	Description
question_id	INT	PK, Auto Increment	Unique identifier of the question.
quiz_id	INT	FK, NOT NULL	References the Quiz containing the question.
content	TEXT	NOT NULL	Content of the question.
question_type	VARCHAR(50)	NOT NULL	Type of the question.

** Answer*

The Answer table stores predefined answers associated with questions.

Column	Data Type	Constraints	Description
answer_id	INT	PK, Auto Increment	Unique identifier of the predefined answer.
question_id	INT	FK, NOT NULL	References the Question associated with the answer.
content	TEXT	NOT NULL	Content of the predefined answer.
is_correct	BOOLEAN	NOT NULL	Indicates whether the answer is considered correct.

** QuizSubmission*

The QuizSubmission table stores information representing a submitted attempt for a quiz.

Column	Data Type	Constraints	Description
submission_id	INT	PK, Auto Increment	Unique identifier of the quiz submission.
quiz_id	INT	FK, NOT NULL	References the Quiz associated with the submission.
submitted_at	DATETIME	NOT NULL	Date and time when the quiz was submitted.

** SubmissionAnswer*

The SubmissionAnswer table stores responses provided for individual questions within a quiz submission.

Column	Data Type	Constraints	Description
submission_answer_id	INT	PK, Auto Increment	Unique identifier of the submitted answer.
submission_id	INT	FK, NOT NULL	References the QuizSubmission containing the response.
question_id	INT	FK, NOT NULL	References the Question being answered.
answer_id	INT	FK, NULL	References a predefined Answer when applicable.
response_text	TEXT	NULL	Stores a textual response when applicable.

2. Use Case Specifications

2.1 UC01 – View Quiz List

Item	Specification
Use Case ID	UC01
Use Case Name	View Quiz List
Primary Actor	Trainer
Secondary Actor	None
Description	As a Trainer, I want to view the existing quizzes so that I can review and manage the quiz content available in the system.
Preconditions	The Trainer can access the Quiz Management function.
Postconditions	The list of existing quizzes is presented to the Trainer. No quiz data is modified.
Normal Flow	1. The Trainer accesses the Quiz Management function. 2. The system retrieves the available quiz data. 3. The system displays the quiz list. 4. The Trainer views the displayed quiz list.
Alternative Flow A1 – No quizzes are available	At Step 2: 1. The system determines that no quizzes are currently available. 2. The system displays an empty quiz list with an appropriate informational message. 3. The use case ends.
Alternative Flow A2 – Quiz data cannot be retrieved	At Step 2: 1. The system encounters an error while retrieving quiz data. 2. The system does not display incomplete or invalid quiz information.

	3. The system displays an appropriate error message to the Trainer. 4. The use case ends.

2.2 UC02 – View Quiz Details

Item	Specification
Use Case ID	UC02
Use Case Name	View Quiz Details
Primary Actor	Trainer
Secondary Actor	None
Description	As a Trainer, I want to view the details of a selected quiz so that I can review its information, questions, and answers before managing its content.
Preconditions	The Trainer can access the Quiz Management function.
Postconditions	The selected quiz information is presented to the Trainer. No quiz data is modified.
Normal Flow	1. The Trainer views the quiz list. 2. The Trainer selects a quiz to view. 3. The system retrieves the selected quiz and its associated questions and answers. 4. The system displays the quiz details to the Trainer. 5. The Trainer reviews the displayed information.
Alternative Flow A1 – Quiz not found	At Step 3: 1. The system cannot find the selected quiz. 2. The system displays an appropriate error message. 3. The use case ends.
Alternative Flow A2 – Quiz data cannot be retrieved	At Step 3: 1. The system encounters an error while retrieving the quiz data. 2. The system displays an appropriate error message. 3. The use case ends.

2.3 UC03 – Create Quiz

Item	Specification
Use Case ID	UC03
Use Case Name	Create Quiz
Primary Actor	Trainer
Secondary Actor	None
Description	As a Trainer, I want to create a new quiz so that I can prepare quiz content for internal training assessments.
Preconditions	The Trainer can access the Quiz Management function.
Postconditions	A new quiz is created and stored in the system. The newly created quiz becomes available for further management.
Normal Flow	1. The Trainer selects the function to create a new quiz. 2. The system provides the required fields for quiz information. 3. The Trainer provides the quiz title, description, and duration. 4. The Trainer confirms the creation request.

	5. The system validates the provided quiz data. 6. The system creates the new quiz. 7. The system confirms that the quiz has been created successfully.
Alternative Flow A1 – Invalid quiz data	At Step 5: 1. The system detects missing or invalid quiz information. 2. The system rejects the creation request. 3. The system identifies the invalid information to the Trainer. 4. The Trainer corrects the information. 5. The flow resumes at Step 4.
Alternative Flow A2 – Quiz cannot be created	At Step 6: 1. The system encounters an error while creating the quiz. 2. The quiz is not added to the system. 3. The system displays an appropriate error message. 4. The use case ends.

2.4 UC04 – Update Quiz

Item	Specification
Use Case ID	UC04
Use Case Name	Update Quiz
Primary Actor	Trainer
Secondary Actor	None
Description	As a Trainer, I want to update an existing quiz so that I can keep its information accurate and appropriate for the training assessment.
Preconditions	The Trainer can access the Quiz Management function. The Trainer has selected a quiz to update.
Postconditions	The selected quiz contains the updated information. Other quizzes remain unchanged.
Normal Flow	1. The Trainer selects an existing quiz to update. 2. The system retrieves and displays the current quiz information. 3. The Trainer modifies the required quiz information. 4. The Trainer confirms the update request. 5. The system validates the modified data. 6. The system applies the changes to the selected quiz. 7. The system confirms that the quiz has been updated successfully.
Alternative Flow A1 – Quiz not found	At Step 2: 1. The system cannot find the selected quiz. 2. The system displays an appropriate error message. 3. No quiz data is modified. 4. The use case ends.
Alternative Flow A2 – Invalid quiz data	At Step 5: 1. The system detects invalid quiz information. 2. The system rejects the update request. 3. The system identifies the invalid information to the Trainer. 4. The Trainer corrects the information. 5. The flow resumes at Step 4.
Alternative Flow	At Step 6: 1. The system encounters an error while applying the changes.

Item	Specification
A3 – Quiz cannot be updated	2. The system does not apply an incomplete update. 3. The system displays an appropriate error message. 4. The use case ends.

2.5 UC05 – Delete Quiz

Item	Specification
Use Case ID	UC05
Use Case Name	Delete Quiz
Primary Actor	Trainer
Secondary Actor	None
Description	As a Trainer, I want to delete an existing quiz so that quiz content that is no longer required can be removed from the system.
Preconditions	The Trainer can access the Quiz Management function. The Trainer has selected a quiz to delete.
Postconditions	The selected quiz is removed from the system. Other quizzes remain unchanged.
Normal Flow	1. The Trainer selects an existing quiz to delete. 2. The system identifies the selected quiz. 3. The system requests confirmation of the deletion. 4. The Trainer confirms the deletion request. 5. The system deletes the selected quiz. 6. The system confirms that the quiz has been deleted successfully.
Alternative Flow A1 – Trainer cancels deletion	At Step 4: 1. The Trainer cancels the deletion request. 2. The system does not modify the quiz data. 3. The use case ends.
Alternative Flow A2 – Quiz not found	At Step 2: 1. The system cannot find the selected quiz. 2. The system displays an appropriate error message. 3. The use case ends.
Alternative Flow A3 – Quiz cannot be deleted	At Step 5: 1. The system encounters an error while deleting the quiz. 2. The system displays an appropriate error message. 3. The use case ends.

2.6 UC06 – Manage Questions

Item	Specification
Use Case ID	UC06
Use Case Name	Manage Questions
Primary Actor	Trainer
Secondary Actor	None
Description	As a Trainer, I want to manage the questions associated with a quiz so that the quiz contains appropriate content for the training assessment.

Item	Specification
Preconditions	The Trainer can access the Quiz Management function. The Trainer has selected a quiz whose questions are to be managed.
Postconditions	The question list of the selected quiz reflects the successfully applied changes, while the relationship between the quiz and its questions remains consistent.
Normal Flow	1. The Trainer selects a quiz whose questions are to be managed. 2. The system retrieves the selected quiz and its associated questions. 3. The system displays the current question information. 4. The Trainer selects an action to add, update, or remove a question. 5. For an add or update operation, the Trainer provides or modifies the question content and question type. 6. The Trainer confirms the requested change. 7. The system validates the question data and requested operation. 8. The system applies the change to the selected quiz. 9. The system confirms that the question information has been updated successfully.
Alternative Flow A1 – Quiz not found	At Step 2: 1. The system cannot find the selected quiz. 2. The system displays an appropriate error message. 3. The use case ends.
Alternative Flow A2 – Invalid question data	At Step 7: 1. The system detects missing or invalid question information. 2. The system rejects the requested change. 3. The system identifies the invalid information to the Trainer. 4. The Trainer corrects the information. 5. The flow resumes at Step 6.
Alternative Flow A3 – Trainer cancels question removal	At Step 6, for a remove operation: 1. The Trainer cancels the removal request. 2. The system keeps the existing question information unchanged. 3. The use case ends.
Alternative Flow A4 – Question change cannot be applied	At Step 8: 1. The system encounters an error while applying the requested change. 2. The system does not leave the quiz-question relationship in an inconsistent state. 3. The system displays an appropriate error message. 4. The use case ends.

2.7 UC07 – Manage Answers

Item	Specification
Use Case ID	UC07
Use Case Name	Manage Answers
Primary Actor	Trainer
Secondary Actor	None
Description	As a Trainer, I want to manage the answers associated with a question so that the question contains the answer information required for the training assessment.

Item	Specification
Preconditions	The Trainer can access the Quiz Management function. The Trainer has selected a question whose answers are to be managed.
Postconditions	The answers associated with the selected question reflect the successfully applied changes, and the relationship between the question and its answers remains consistent.
Normal Flow	1. The Trainer selects a question whose answers are to be managed. 2. The system retrieves the selected question and its associated answers. 3. The system displays the current answer information. 4. The Trainer selects an action to add, update, or remove an answer. 5. For an add or update operation, the Trainer provides or modifies the answer information and specifies whether the answer is correct where applicable. 6. The Trainer confirms the requested change. 7. The system validates the answer data and requested operation. 8. The system applies the change to the selected question. 9. The system confirms that the answer information has been updated successfully.
Alternative Flow A1 – Question not found	At Step 2: 1. The system cannot find the selected question. 2. The system displays an appropriate error message. 3. The use case ends.
Alternative Flow A2 – Invalid answer data	At Step 7: 1. The system detects missing or invalid answer information. 2. The system rejects the requested change. 3. The system identifies the invalid information to the Trainer. 4. The Trainer corrects the information. 5. The flow resumes at Step 6.
Alternative Flow A3 – Trainer cancels answer removal	At Step 6, for a remove operation: 1. The Trainer cancels the removal request. 2. The system keeps the existing answer information unchanged. 3. The use case ends.
Alternative Flow A4 – Answer change cannot be applied	At Step 8: 1. The system encounters an error while applying the requested change. 2. The system does not leave the question-answer relationship in an inconsistent state. 3. The system displays an appropriate error message. 4. The use case ends.

2.8 UC08 – View Available Quizzes

Item	Specification
Use Case ID	UC08
Use Case Name	View Available Quizzes
Primary Actor	Trainee
Secondary Actor	None

Item	Specification
Description	As a Trainee, I want to view the available quizzes so that I can select a quiz to participate in as part of my training.
Preconditions	The Trainee can access the Quiz Participation function.
Postconditions	The available quizzes are presented to the Trainee. No quiz data is modified.
Normal Flow	1. The Trainee accesses the Quiz Participation function. 2. The system retrieves the quizzes available for participation. 3. The system displays the available quiz list. 4. The Trainee views the displayed quizzes.
Alternative Flow A1 – No quizzes are available	At Step 2: 1. The system determines that no quizzes are currently available for participation. 2. The system displays an empty quiz list with an appropriate informational message. 3. The use case ends.
Alternative Flow A2 – Quiz data cannot be retrieved	At Step 2: 1. The system encounters an error while retrieving quiz data. 2. The system does not display incomplete or invalid quiz information. 3. The system displays an appropriate error message to the Trainee. 4. The use case ends.

2.9 UC09 – View Quiz Details

Item	Specification
Use Case ID	UC09
Use Case Name	View Quiz Details
Primary Actor	Trainee
Secondary Actor	None
Description	As a Trainee, I want to view the general information of a selected quiz so that I can understand the assessment before deciding to start it.
Preconditions	The Trainee can access the Quiz Participation function. The Trainee has selected a quiz from the available quiz list.
Postconditions	The general information of the selected quiz is presented to the Trainee. No quiz data is modified.
Normal Flow	1. The Trainee views the available quiz list. 2. The Trainee selects a quiz to view. 3. The system retrieves the selected quiz information. 4. The system displays the quiz title, description, and duration to the Trainee. 5. The Trainee reviews the displayed quiz information.
Alternative Flow A1 – Quiz not found or unavailable	At Step 3: 1. The system cannot find the selected quiz or determines that it is no longer available for participation. 2. The system displays an appropriate informational or error

Item	Specification
	message. 3. The use case ends.
Alternative Flow A2 – Quiz information cannot be retrieved	At Step 3: 1. The system encounters an error while retrieving the selected quiz information. 2. The system displays an appropriate error message. 3. The use case ends.

2.10 UC10 – Take Quiz

Item	Specification
Use Case ID	UC10
Use Case Name	Take Quiz
Primary Actor	Trainee
Secondary Actor	None
Description	As a Trainee, I want to take an available quiz so that I can complete a knowledge assessment as part of my training.
Preconditions	The Trainee can access the Quiz Participation function. The Trainee has selected a quiz to start.
Postconditions	The Trainee's valid quiz responses are accepted by the system, and the quiz submission is completed successfully.
Normal Flow	1. The Trainee selects an available quiz to start. 2. The system retrieves the selected quiz and its questions. 3. The system presents the quiz questions and appropriate answer inputs according to each question type. 4. The Trainee provides answers to the quiz questions. 5. The Trainee requests to submit the completed quiz. 6. The system validates the submitted responses. 7. The system accepts the valid quiz responses. 8. The system confirms that the quiz has been submitted successfully. 9. The use case ends.
Alternative Flow A1 – Quiz not found or unavailable	At Step 2: 1. The system cannot find the selected quiz or determines that the quiz is no longer available. 2. The system does not start the quiz. 3. The system displays an appropriate informational or error message. 4. The use case ends.
Alternative Flow A2 – Quiz content cannot be retrieved	At Step 2: 1. The system encounters an error while retrieving the quiz content. 2. The system does not present incomplete quiz content to the Trainee. 3. The system displays an appropriate error message. 4. The use case ends.
Alternative Flow	At Step 6: 1. The system detects invalid or inconsistent response data.

A3 – Invalid submission data	2. The system rejects the submission. 3. The system informs the Trainee that the submitted responses cannot be accepted. 4. The Trainee corrects the applicable responses when possible. 5. The flow resumes at Step 5.
Alternative Flow A4 – Submission cannot be processed	At Step 7: 1. The system encounters an error while processing the submission. 2. The system does not report the quiz as successfully submitted. 3. The system displays an appropriate error message. 4. The use case ends.